INVENTION/TECHNOLOGY/INNOVATION/IDEA DISCLOSURE FORM

WAYNE STATE UNIVERSITY

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Date Rec'd.: _____

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Grant No.: _____

Federal funding?_____

The purpose of this disclosure is to document and identify new innovations, inventions, technology, or innovative business ideas developed by members of the Wayne State community including faculty and staff. This form is an important document, and is intended to help us meet our government reporting obligations and to provide a means of determining commercial potential and assessing intellectual property protection.

Submit this form directly to InventionDisclosures@wayne.edu. It will be acknowledged with a received date **only when submitted via the link above. **Please do not submit this disclosure without all required signatures (page 6).**

Verify that the Memorandum of Understanding document (pages 8-10) and the Royalty Division Agreement (page 11) have been completed prior to submitting document.

1. **Title of Invention/Technology/Innovation/Idea**

Automated detection of transitional vertebral anatomy for wrong-level spine surgery prevention: (A) a segmenter-agnostic classifier that flags transitional vertebrae on preoperative CT, and (B) a neural network that names vertebral levels from a single intraoperative C-arm exposure without registration to a preoperative scan

2. **Contact Information**

Please include each submitter substantively involved with the invention/technology/innovation/idea and a brief description of their role. Final inventors for any patent will be determined by the law firm at the time of patent filing. Fill in the name in appropriate table below with a few words regarding their role in the invention/technology/innovation/idea and complete the detailed contact information on the last two pages of this document.

(a) **WSU** affiliated submitters:

SUBMITTER NAME	ROLE IN THE DEVELOPMENT OF THE INVENTION/TECHNOLOGY/INNOVATION/IDEA
Gregory John Schwing, MD, PhD	Conceived both parts; designed and trained the networks; built the training corpus (rendered CT + real C-arm frames), the per-device image-formation models, the evaluation protocol and the software; sole inventor.

(b) **Non- WSU** affiliated submitters:

SUBMITTER NAME	F ROLE IN THE DEVELOPMENT OF THE INVENTION/TECHNOLOGY/INNOVATION/IDEA
None. The sole inventor is the WSU-affiliated submitter above; co-authors of the related papers contributed data annotation, manuscript review or mentorship, not to the inventive concept.	

3. **External Collaborations**

Do any WSU affiliated submitters have joint appointments or other external affiliations with an obligation to assign or disclose inventions?	X	Yes		No
If yes, please list external Institutions (example VA)	Detroit Medical Center (employer of record: Detroit Education and Research; Tenet Healthcare Corporation): the inventor is a resident physician in the Department of Surgery under a Graduate Medical Education Agreement (effective 1 July 2026). That agreement contains no intellectual-property or invention-assignment clause; its section 10 is a confidentiality clause covering the agreement's terms and proprietary information provided by the employer. No DMC resource, data or patient information was used for the invention.			
Please indicate the relative percent (%) contribution between WSU and non-WSU parties.	100	WSU%	0	Non-WSU%

DESCRIPTION OF INVENTION/TECHNOLOGY/INNOVATION/IDEA

4. **Provide a narrative description of the invention/technology/innovation/idea. What is novel or unexpected? Please provide sufficient technical detail to convey a clear understanding to the extent known at the time of disclosure.**

Problem. Wrong-level spine surgery is a never event whose largest anatomic driver is a transitional vertebra (a sixth lumbar vertebra, a thirteenth thoracic vertebra, or a fifth lumbar vertebra fused to the sacrum; 5-35 % of people). Planning is done on lumbar-only CT without a count from C2, and intraoperative confirmation is counting vertebral bodies on fluoroscopy; both fail on transitional spines, and registration-based aids (LevelCheck-type systems) project the labels of the preoperative read and inherit its counting error. Current CT segmenters (TotalSegmentator, SPINEPS) shift the lumbar labels one level on such spines without warning.

Part A -- preoperative CT flag (near-term product). A classifier that takes the output of ANY CT segmenter, discards the segmenter's level names, and computes name-free per-vertebra features (rib presence traced in the CT image, rib-free body count above the sacral plateau, transverse-process-to-sacral-ala contact area, disc and end-plate geometry, iliac crest line), then flags the transitional phenotype (L6, sacralised L5 by Castellvi grade, T13, lumbar rib, four lumbar) with a plain-language reason, an abstention when the junction is outside the field of view, and an 'unusual spine' score.

Part B -- intraoperative level naming. A convolutional-recurrent network reads a single C-arm exposure, detects the vertebral bodies and the sacrum, and names each level from LOCAL MORPHOLOGY rather than from a count, so a transitional vertebra is recognised as what it is; the classifier is coupled with a count from the detected sacrum through a set of candidate vertebral sequences ('alphabets', ordinary and transitional) and abstains visibly when they disagree. No registration, fiducials or navigation hardware. It is trained on digitally reconstructed radiographs rendered from labelled CT through per-device image-formation models (fitted physics parameters per C-arm model; free geometric labels by projection, including the bicoxofemoral point) plus few real frames per device. Measured: on a C-arm never seen in training it named 96.4 % of detected lateral vertebrae zero-shot (595 of 617) and 97.4 % after case-grouped fine-tuning; in its feature space a sixth lumbar vertebra separates from the fifth more than the fifth from the fourth (silhouette 0.35 vs 0.23).

Novel or unexpected: level identity learned from local morphology instead of counting or registration; the alphabet coupling with visible abstention; per-device image-formation models that let one network serve several C-arms from renders with few real frames; a segmenter-agnostic, name-free CT flag. Unexpected: naming accuracy on an unseen C-arm reached its plateau before any real frame of that device was used for training -- the limit is coverage of acquisition geometry, not the number of real images.

(a) Does the invention/technology/innovation/idea involve a new process or method, composition of matter or device, software or combination?

Combination: new processes/methods (name-free transitional-phenotype classification from segmentations; level naming from local morphology coupled with a count and abstention; render-based per-device training) embodied in software (trained neural-network weights, training and inference code, per-device image-formation parameters).

(b) Does the invention/technology/innovation/idea involve a new use, or an improvement of an existing product or process?

Improvement of existing processes: preoperative CT planning (a flag on the planning scan) and intraoperative fluoroscopic level confirmation. New use of any vendor's CT segmenter output as the input of a transitional-anatomy flag.

(c) Describe the product/services that are expected from the development of the invention/technology/innovation/idea.

(A) Software as a medical device that flags suspected transitional anatomy on preoperative CT and notifies the reader -- radiology/PACS workflow, Class II 510(k) anticipated (product codes QIH/MYN), the near-term product. (B) Intraoperative level confirmation on the C-arm exposure -- decision support beside the surgeon's count, integrated with C-arm consoles or navigation platforms (harder regulatory path; later). (C) Licensable assets: per-device image-formation parameters, the evaluation protocol and reference standard, trained weights retrained on commercially licensed data.

5. **Give the approximate date or period of time when the invention/technology/innovation/idea was conceived.**

Part A (name-free transitional-phenotype classifier on CT segmentations): the CT-side count-anomaly work began spring 2026; the name-free, segmenter-agnostic classifier as claimed was designed on 24-25 September 2026. Part B (level naming from local morphology on a single C-arm exposure): 24 August 2026; the alphabet coupling and abstention: September 2026; per-device image-formation models: 21-23 September 2026.

6. **Please provide the location and type of any University laboratory records and/or data available related to the invention/technology/innovation/idea. Give reference numbers and physical locations, but do not enclose.**

Version-controlled repositories with dated commits: spinesurg-ct-nnunet (private, GitHub: Part B, the CT classifier, training and evaluation pipelines), CTSpinoPelvic1K (public) and OpenSpineToolkit (public); the WSU Grid high-performance computing account (job records, trained checkpoints, model registry, results under /rs/rs_grp_oschome/go2432); Zenodo deposits of the dataset (concept DOI 10.5281/zenodo.22139642); the inventor's dated laboratory journal (PDF); the manuscripts in preparation with their result ledgers.

7. **What is the stage of development of the invention/technology/innovation/idea? E.g., still in conceptual/idea stage, functioning prototype, fully developed product? Please explain.**

Functioning research prototypes: trained weights exist for Part B and are evaluated on public intraoperative datasets (C-ArmLSpine C-arm frames with cone-beam CT truth; COBRA2026 kV frames) and on rendered CT; Part A's feature extractor runs on 3,500 CT-segmenter pairs and its classifier is in training; a manuscript on Part B is in preparation; no prospective or clinical evaluation; no regulatory submission. The prototypes are trained on non-commercially licensed research data (see Resources); a commercial model must be rebuilt from commercially licensed data.

COMMERCIAL OPPORTUNITY

8. **What is the market opportunity for the invention/technology/innovation/idea? Please describe the market, potential customers, and unmet need or problem/solution it addresses.**

Every spine operation confirms its level under fluoroscopy and is planned on CT or MRI; over one million spine procedures are performed each year in the United States. Wrong-level surgery is a reportable never event (half of responding spine surgeons report having performed one; registry rate about 0.1 %, higher in the thoracic spine), and transitional vertebrae (5-35 % prevalence) are its largest anatomic cause. Customers: radiology groups and PACS/AI platform vendors (Part A); hospitals and ambulatory surgery centres, C-arm and navigation manufacturers (Part B). Unmet need: a check that does not fail on exactly the anatomy that causes the error, works on the limited fields of view actually acquired, and needs no registered preoperative scan.

(a) How is the need currently being met and/or any known alternative solutions or competitive products? Describe the commercial market in which the invention/technology/innovation/idea would compete.

Manual counting on fluoroscopy with a marker and a check film; intraoperative navigation (registration of the preoperative CT/CBCT to the patient; hardware, fiducials, time); LevelCheck-type research systems (Johns Hopkins) that register the preoperative scan and project its labels -- accurate on ordinary spines, dependent on the planning read; CT segmenters and vendor spine-labelling tools (TotalSegmentator/3D Slicer, MSKai-type products) that mislabel transitional spines by one level without warning. No product today flags a transitional vertebra on the planning CT or names one from a single C-arm exposure.

(b) Please identify major industry players (if known).

Medtronic (StealthStation, Mazor), Globus Medical (ExcelsiusGPS), Stryker, Brainlab, Siemens Healthineers (Cios), GE HealthCare (OEC), Ziehm Imaging, Philips; radiology AI platforms (Aidoc, Viz.ai, Avicenna CINA, Nuance/Microsoft); open-source segmenters (TotalSegmentator/3D Slicer, SPINEPS).

(c) What advantages does the invention/technology/innovation/idea possess over existing or emerging technologies?

Built for the transitional anatomy on which counting, registration and current segmenters fail; segmenter-agnostic and name-free, so it audits any vendor's output without retraining it; registration-free intraoperative naming from one exposure with visible abstention; one network serves several C-arm models through per-device image-formation parameters, so adapting to a new device needs few real frames; every call carries a plain-language reason.

(d) Has any commercial interest been shown to date? Please Identify companies, with contact name, email, or phone number, if known to you.

No commercial interest has been solicited or received to date.

(e) Are you interested in forming, or have you formed, a company to commercialize this technology?

Interested in forming a company to commercialise the technology; none formed.

INVENTION DISCLOSURE

9. **Public Disclosure**

"Published" or "publicly disclosed" means dissemination of information to members of the public about the invention, or otherwise making such information available to or accessible by members of the public. It Includes publications, oral presentations, posters, meeting abstracts, reports, public lectures, new releases, distribution to commercial company (or companies) without restriction, Internet and social media posts, online databases, etc**.** "Publicaly disclosed" also includes releases via an app store or other download website**.**

Has the invention/technology/innovation/idea been published or publicly disclosed?	X Yes ☐ No
(a) If yes, please state the date, place and circumstances of any public disclosures pertaining to the invention/technology.	Partial. Neither the name-free CT classifier (Part A as claimed) nor Part B (level naming from a single C-arm exposure by local morphology, the alphabet coupling, the per-device image-formation models) has been publicly disclosed; Part B's manuscript is in preparation. Disclosed so far, all research-track: (a) 'Not All Spines Are Created Equal: How CT Segmenters Fail on Lumbosacral Transitional Vertebrae', NeurIPS 2026 Evaluations and Datasets Track, submission 3474, accepted 24 Sep 2026 (OpenReview D8yRWDAGQF, public): the TotalSegmentator one-level label shift on transitional spines, the dataset, and a merge-based CT segmenter released with weights (transitional-aware oversampling is used there). (b) Abstracts accepted to the 2026 Congress of Neurological Surgeons annual meeting (Washington, DC), including the single-shot CT detection of count anomalies ('Not All Spines Are Created Equal', mini-oral, scheduled 2 November 2026; slides not yet submitted; accepted abstracts are published in the meeting's abstract supplement). (c) 'Harnessing Algorithmic Disagreement for Automated Lumbosacral Transitional Vertebrae Detection', Tech Fair presentation and hands-on demonstration, 43rd AACA annual meeting, Mayo Clinic, Rochester MN, Saturday 13 June 2026 (slides attached): a separate, earlier project -- an LSTV screening pipeline assembled from third-party open-source segmenters (TotalSpineSeg, SPINEPS/VERIDAH) whose level disagreement flags the phenotype, with Castellvi heuristics -- released as open source. It is listed for completeness; it does not disclose Part A's name-free classifier or Part B, neither of which existed then, and neither claim relies on that mechanism. (d) CTSpinoPelvic1K dataset: public Zenodo releases (v10 on 7 Sep 2026, v11 on 18 Sep 2026); arXiv preprint 2609.22760 (19 Sep 2026) of a dataset article under review at Medical Physics -- the CT resource and its label scheme, not a method. (e) openspineconsortium.com: public website with a spinopelvic-measurement demonstration and the open OpenSpineToolkit; neither the CT classifier nor the level-naming network is served there.
(b) If no, please provide any information regarding future public disclosure plans (possible dates).	Planned: (1) the Part B manuscript (Radiology: Artificial Intelligence) is complete and is being HELD from submission until this office advises -- it is the first disclosure of Part B (a copy accompanies this form); (2) the NeurIPS 2026 camera-ready will be published in the proceedings about one month before the December conference (~November 2026) -- dataset and segmenter audit, as in (a); (3) CNS

	2026 mini-oral, 2 November 2026, CT-side count anomalies; (4) the dataset article (Medical Physics) is being submitted -- dataset only, no method of this disclosure.
Pertaining to software, are there any current or future plans to publish in a journal that requires Open Source?	X Yes ☐ No
If yes, please state the date, place and circumstances of the disclosure.	The NeurIPS 2026 paper released its CT segmenter code and weights under an open licence (public GitHub OpenSpineConsortium/spinopelvic-seg; Hugging Face) -- that segmenter is not the claimed invention. For Part A's classifier and Part B's network no open-source release of weights is planned before the office's assessment; the journal targeted (Radiology: AI) expects code availability, which can be met with a gated or on-request release.

10. **Manuscripts/Public Use**

Do you have a draft manuscript?	X Yes	☐ No
Has a manuscript been submitted for publication?	X Yes	☐ No
Has a manuscript been accepted as of this date?	X Yes	☐ No
Has a manuscript published?	X Yes	☐ No
If yes, provide details if not noted above.	Draft: the Part B manuscript, unsubmitted. Submitted: the dataset article (Medical Physics,	
Is there any planned offer for sale, sale or public use of the invention/technology/innovation/idea?	☐ Yes	X No
If yes, please provide additional detail (including dates).	None planned before assessment.	
Has the invention/technology/innovation/idea already been offered for sale, sold or used publically?	☐ Yes	X No
If yes, please provide additional detail (including dates).	Not offered for sale, sold or used publicly.	

FUNDING & RESOURCES

11. **Funding**

Was the work that led to the invention/technology/innovation/idea sponsored or otherwise funded or supported in whole or in part by another organization or the government?	X Yes	☐ No

If Yes, please identify in the blanks below, all sources of public or private funding (e.g., contracts or grants from federal or state agencies, foundations, public or private organizations or industry).

(a) **Title of Government Agency**: National Institutes of Health, National Institute of General Medical Sciences

Grant/Contract No.: 5F30GM147909 (Ruth L. Kirschstein NRSA individual predoctoral MD/PhD fellowship; ended 31 May 2026)

Principal Investigator: _____ Phone: _____

Gregory John Schwing (fellow, PI of record) 504-994-0403

(b) **Name of Industrial Sponsor**: None

Grant/Contract No.: _____

Principal Investigator: _____ Phone: _____

(c) **Other sponsor(s):** Wayne State University Grid high-performance computing (compute and storage allocation); no other sponsor

Grant/Contract No.: _____

Principal Investigator: _____ Phone: _____

12. **Resources**

Did you use proprietary materials owned by another organization to make this invention/technology or under for example any materials shared under a Material Transfer Agreement (MTA) or data use agreement? [X] Yes [ ] No

**Examples of proprietary materials: confidential information, biological materials such as cell lines, transgenic animals, vectors, or genetic sequences, chemical compounds, data and software or computer source code.*

If yes, please attach a copy of the documentation or agreement, which covered the transfer of such material to you or the University. Yes. (i) VinDr-SpineXR radiographs were obtained under the PhysioNet Restricted Health Data Use Agreement 1.5.0 (research use only) and were used to train evaluation fold models only. (ii) The research prototype's training data carry non-commercial licences: CTSpinoPelvic1K inherits CC BY-NC-SA 4.0 from CTSpine1K; COBRA2026 CC BY-NC 4.0; DeepFluoro CC BY-NC-SA 4.0; Ljubljana CC BY-NC-ND; VerSe share-alike; permissive: C-ArmLSpine CC BY 4.0, TCIA CT Colonography, NHANES II (public domain), TotalSegmentator dataset CC BY 4.0. Any commercial model must be retrained from commercially licensed data; the research weights are not the commercial asset. Copies of the PhysioNet agreement and the licences can be attached.

During the course of your work on this invention/technology, were you or any of the WSU investigators listed above salaried with, or employed by, any organization or entity that may have a right to this technology? [X] Yes [ ] No

If yes, please attach a statement of particulars. Yes: the lead inventor is a salaried resident physician of the Detroit Medical Center (Tenet Healthcare Corporation), Department of Surgery, from 1 July 2026 under a Graduate Medical Education Agreement (the earliest work, April-June 2026, preceded that appointment, during the final months of medical school and the F30 fellowship). The agreement carries no invention-assignment clause.

13. Please specify the nature of University Support (including salary, funds, facilities, services and/or equipment. Identify any applicable internal WSU funding programs (e.g. TDI, MTRAC) that have supported the invention/technology. **WSU Grid high-performance computing (GPU and CPU allocations, CephFS storage), WSU library access, faculty sponsorship (Department of Computer Science). The inventor performed all of the work himself; WSU's contribution was compute and storage on the Grid. No WSU salary support; no TDI/MTRAC funding.**

SIGNATURES

WSU Faculty and other Employees, please do not submit this disclosure without department/dean signatures as appropriate or without the signatures of all persons named as WSU submitters in item 2.

NAMES	SIGNATURES	DATE:
SUBMITTER		
SUBMITTER		
SUBMITTER		
SUBMITTER		
CHAIR OR DEPARTMENT HEAD(Need all if multiple departments)		
CHAIR OR DEPARTMENT HEAD		

DEAN or VP (Need all if multiple departments)

DEAN

SUBMITTER INFORMATION

To facilitate potential payments to investigators, maintenance of the Technology Transfer Office database and to ensure the accuracy of legal documents, please supply the following information.
**Home address and citizenship is necessary to comply with certain patent office requirements.*

1. Name Gregory John Schwing Citizenship: United States (born in the U.S.) If not U.S. citizen
   Indicate visa type: not applicable (U.S. citizen)
   and check the following: ____Nonresident alien ____Resident alien
   Job Title or Resident physician, General Surgery (Detroit Medical Center / Wayne State University School of Medicine)
   Student Status Resident physician (WSU School of Medicine affiliate; employed by DMC) Degree MD, PhD (Wayne State University, May 2026)
   Department Surgery College/Division School of Medicine
   WSU Address Department of Surgery, Wayne State University School of Medicine, Detroit, MI 48201
   Phone 504-994-0403
   Home Address 206 Josephine Ave, Royal Oak, MI 48067-2217, United States Phone 504-994-0403
   E-mail gregory.schwing@med.wayne.edu Optional: Gender_____

2. Name _____ Citizenship: _____ If not U.S. citizen
   Indicate visa type: _____
   and check the following: ____Nonresident alien ____Resident alien
   Job Title or
   Student Status _____ Degree _____
   Department _____ College/Division _____
   WSU Address _____ Phone _____
   Home Address _____ Phone _____
   E-mail _____ Optional: Gender_____

3. Name _____ Citizenship: _____ If not U.S. citizen
   Indicate visa type: _____
   and check the following: ____Nonresident alien ____Resident alien
   Job Title or
   Student Status _____ Degree _____
   Department _____ College/Division _____
   WSU Address _____ Phone _____
   Home Address _____ Phone _____
   E-mail _____ Optional: Gender_____

4. Name _____ Citizenship: _____ If not U.S. citizen
   Indicate visa type: _____
   and check the following: ____Nonresident alien ____Resident alien
   Job Title or
   Student Status _____ Degree _____
   Department _____ College/Division _____
   WSU Address _____ Phone _____
   Home Address _____ Phone _____
   E-mail _____ Optional: Gender_____

NON-WSU SUBMITTER INFORMATION

Investigator(s) Name	**Home Address**	**Email**	**Citizenship**	**Phone No.**	**Non-WSU Institution/Contact No.**

MEMORANDUM OF UNDERSTANDING

WHEREAS, the Wayne State University (herein "University") Board of Governors' (herein "Board") July 11, 1984, Patent Policy Statute provides authority for establishing relations between the University and University inventors, and

WHEREAS, **INVENTORS** (herein "Inventors"), seeks the University's support in assessing patentability and if warranted filing a patent application(s) for their Invention(s) developed with use of the University's facilities and/or other University support, and

WHEREAS, the Inventors are mindful of the need to make arrangements for the assessment and filing of a patent application(s) for the subject invention(s), in order to secure patent rights, if appropriate, on a timely basis, and

WHEREAS, the University's chief executive officer, in whom the Board has vested authority and responsibility for the University's general management and operations, has delegated to the undersigned representative of the University's Technology Commercialization group the authority to act in his stead in matters relating to the administration of patents, and

WHEREAS, the University and the Inventors recognize the necessity to act without undue delay in assessing and potentially securing patent protection in these regards, and

WHEREAS, the interests of the parties are enhanced by adoption of this agreement,

NOW, THEREFORE, the parties hereto agree that the Inventors hereby assign to the University all their right, title, and interest in and to the subject invention(s) and all corresponding right, title, and interest in intellectual property including patent(s) which may issue with respect to same, and concerning which the University has undertaken the obligation to meet administrative and legal costs; and the parties agree that the University will authorize the expenditure of University resources reasonably necessary to meet the legal and administrative costs and expenses of managing the subject matter of this agreement;

1. Any funds, including, but not limited to, royalties, licensure fees, and proceeds of sale, which are paid to the University by virtue of its proprietary interest in the subject intellectual property, will be distributed so as to reimburse the University for all costs associated with patent assessment, development, application, and licensing, with the remainder of any such funds to be divided between the University and the Inventors according to the following formula:

NET INCOME	INVENTORS	DEPARTMENT	UNIVERSITY
Up to $10,000	75%	15%	10%
Next $90,000	50%	15%	35%
Next $900,000	40%	15%	45%

Over $1 Million	35%	15%	50%

2. All funds to which the Inventors will be entitled, pursuant to the provisions of this Memorandum of Understanding, will be distributed to them at the next bi-annual royalty distribution by the University based on the date of receipt thereof by the University;

3. The University will provide the Inventors with an annual accounting, and the Inventors may have access to examine the records of the subject account during normal business hours;

4. The undersigned representative for the University's Technology Commercialization group has present authority to commit University resources necessary to assess the Invention and, if warranted, secure the subject patent, and said commitment is hereby made;

5. The parties agree to act in good faith, mindful of their mutual and respective interest, and carry out the provisions of this Memorandum of Understanding;

6. As holder of exclusive rights in the subject invention, the University will exercise reasonable judgment in the management of said invention and patent rights pertaining thereto, and the University will make reasonable efforts to consult the Inventors concerning events and decisions that impact on the subject matter of this agreement;

7. Failure of the University to expend reasonable efforts to license or otherwise secure commercial benefits with respect to any invention subject to this agreement within twenty-four (24) months following issuance of any corresponding patent will result in transfer of ownership of the invention to the Inventors; and,

8. The provisions of the Wayne State University Patent Policy Statute, adopted July 11, 1984, are adopted in full as though set forth herein, and said provisions are considered binding on the parties, other provisions of this Memorandum notwithstanding.

FURTHER, the University administration will use its best efforts to secure whatsoever Board authorization and approval is necessary for the University to carry out the terms of this Memorandum of Understanding.

Wayne State University

Date: _____ By: _____

Taunya A. Phillips
Assistant Vice President
Office for Technology Commercialization

_____
Witness

Date: _____ _____

Inventor

_____
Witness

Date: _____ _____

Inventor

_____

Witness

Date: _____ _____

Inventor

_____

Witness

Date: _____ _____

Inventor

_____

Witness

WSU ROYALTY DIVISION AGREEMENT

The undersigned, WSU employees who participated in this technology development**,** do hereby agree to the following division of royalty distribution as between and among ourselves, after having made reimbursement payments for costs and expenses consistent with the Wayne State University Patent Policy and proportionate to our respective contributions:

Gregory John Schwing (sole inventor)	100 %
	%
	%
	%
TOTAL	**100%**

IN WITNESS WHEREOF, we have here to set our hand and seal on this _____ day of **_____, 2026.**

_____ _____
Name Witness

_____ _____
Name Witness

_____ _____
Name Witness

_____ _____
Name Witness

_____ _____
Name Witness

WSU File No.: